Wenqi Lin

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EDUCATION

University of California, Los Angeles (UCLA), *California* *Graduate June 2026 with* Bachelor of Statistics and Data Science

DATA SKILLS

Programming: Python • SQL •R • C++

Analysis: A/B Testing, Prompt Engineering, LLM Evaluation, Regression Analysis, Monte Carlo Simulations, Time-Series Analysis, Experimental Design, Data Analysis and Regression

Tools & Vis: Streamlit, Tableau, Excel, Git, Jupyter Notebooks, Advanced Word, Excel, and PowerPoint skills

EXPERIENCE

Yale | Data Science Intern — Epidemiological Modeling, Yale University Jul 2023 – Sep 2023

- Processed epidemiological datasets and literature-derived parameters with **Dr. Andrea Ayala** to simulate **avian influenza transmission scenarios** using **R (deSolve, tidyverse)**.
- Implemented and compared **SIR and SEIR compartmental models**, performing sensitivity analysis on infection and recovery rates to evaluate outbreak dynamics.
- Conducted **data validation and assumption checks** to ensure robustness of model inputs across multiple case study sources.
- Produced **time-series visualizations and summary reports** to communicate infection peak timing, spread velocity, and recovery patterns to research stakeholders.

Projects

Kairo | Founder | https://meetkairo.app/ July 2026 – Present

- Led **end-to-end product strategy**, roadmap, user research, feature prioritization, and **MVP development**.
- Built the **iOS app**, **backend infrastructure**, **APIs**, authentication, **databases**, and **Firebase services**.
- Developed AI features using **LLMs**, **prompt engineering**, recommendation systems, and workflow automation.
- Designed scalable system architecture and technical infrastructure while iterating rapidly with AI coding tools.
- Led branding, marketing, beta planning, product validation, and go-to-market strategy.

DATA PROJECTS

UCLA Course Enrollment Optimization Project — UCLA, California *3/2026*

R (dplyr, lubridate, ggplot2), Regression Modeling | 33K+ Records

- Cleaned and transformed **33,000+ longitudinal enrollment snapshot records**, resolving duplicate entries, inconsistent timestamps, and dynamic capacity changes.
- Engineered demand features including **fill rate, days-to-capacity, early enrollment velocity, and waitlist pressure metrics**.
- Built **logistic regression models** to predict course fill probability and **Poisson regression models** to model enrollment velocity for count-based demand dynamics.
- Evaluated model performance using **cross-validation, ROC-AUC, and residual diagnostics**, and refined feature sets to improve generalization. And identified that **General Education courses drive 87% higher early demand**, proposing data-driven classroom allocation and scheduling improvements.

Bayesian Analysis of Stock Returns (Apple Inc.) — UCLA, California *8/2024*

Python (NumPy, pandas, matplotlib), MCMC Simulation

- Modeled daily stock return uncertainty using a **Metropolis–Hastings MCMC algorithm (20,000 iterations)** to estimate posterior distributions under different prior assumptions.
- Performed **convergence diagnostics (trace plots, autocorrelation decay)** and posterior predictive checks to validate sampling stability. And Compared informative vs non-informative priors to analyze **risk sensitivity and predictive uncertainty** in financial return modeling.

Islands Cognitive Performance Study - *UCLA, California* *4/2025*

- Designed and ran a **randomized complete block experiment** to assess how 250 mL beverages of varying ABV affect short-term attention.
- Used **R** for data simulation and sampling, power analysis, **one-way ANOVA** with age blocking, residual diagnostics, and post-hoc pairwise tests (**Tukey HSD, Fisher LSD**).
- **Visualized results** and demonstrated that high-ABV beverages (red wine) significantly impair cognitive performance.

Housing Price Regression Analysis - *UCLA, California* *4/2025*

R, Predictive Modeling

- Built a **multiple linear regression model to predict real estate prices**, utilizing **backward stepwise selection (AIC)** to optimize feature selection. And applied **Box-Cox transformations** to correct non-linearity and heteroscedasticity, improving model accuracy and residual normality.